Underground Storage

Depleted Reservoir and Salt Cavern

Design Basis Accident — Military Action

DBA-GAS-FC3 — Facility Class Document

Issued Under: DBA-GAS Sector Framework

Version 0.1 — DRAFT

June 2026

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Prepared for: Natural Gas Sector — Industry, Government, and Allied Partners

Parent document: DBA-GAS Sector Framework v1.4

Preface

Natural gas underground storage is the system's memory. It absorbs gas during low-demand periods and releases it during high-demand periods, buffering the mismatch between the steady-state production and transmission system and the highly variable demand of a winter heating market or a hot-summer generation surge. Without that buffer, the transmission system must match production to demand in real time — an engineering constraint that production and pipeline infrastructure cannot meet during demand peaks. The storage field is therefore not a secondary infrastructure node. It is the mechanism by which a supply system designed for average conditions survives extreme ones.

This document establishes the Design Basis Accident scenarios for deliberate military action against underground natural gas storage infrastructure. The facility class encompasses depleted reservoir storage fields, salt cavern storage fields, and aquifer storage fields, together with the surface infrastructure — wellheads, compression equipment, dehydration systems, and SCADA — that make the subsurface reservoir accessible to the transmission system. The primary reference environments are the Gulf Coast salt cavern complex, which provides the U.S. transmission system's highest-deliverability sprint capacity, and the Appalachian and Midcontinent depleted reservoir fields, which provide the endurance capacity that sustains supply across a full winter season.

FC3 carries an analytical contribution unique in the DBA-GAS series: the inventory depletion attack. Every other facility class in the series produces consequences through physical destruction — a compressor station is blown up, a processing plant is set on fire, a distribution system is severed. The storage field presents an adversary with a different option: deplete the inventory before the demand event that the storage was built to buffer, using compromised withdrawal scheduling systems to drain the reservoir at rates that appear operationally normal until the inventory is exhausted. No physical damage is required. No attack signature is visible on the surface. The consequence — a transmission system without buffer capacity entering peak demand season — arrives weeks after the attack has been completed and attributed, when the remediation window has closed.

The evidentiary anchor for the non-adversarial Design Basis is Aliso Canyon (2015-2016): a single storage well casing failure at the Southern California Gas Company's largest storage field produced an eleven-week blowout, the loss of approximately 5 billion cubic feet of working gas inventory, and supply curtailment that required emergency regulatory action and demand-side management across the Southern California market during a winter heating period. The DBA-MA extension asks what that consequence looks like when the initiating condition is not a corroded casing failing over years, but a deliberate adversarial action designed to produce the same well integrity failure at a time and place of the adversary's choosing.

1. Introduction

1.1 Purpose

This document establishes the Design Basis Accident — Military Action scenarios for the underground natural gas storage facility class. Its purpose is to define the bounding threat event that storage field security planning, resilience investments, and recovery architecture must address. As with all DBA-MA documents, this document does not prescribe specific protective measures at individual facilities. It defines the consequence envelope within which protective measures must perform.

A storage field security program designed against conventional physical threats will fail when a sophisticated adversary combines SCADA pre-positioning against the field's well management and withdrawal scheduling systems with selective kinetic action against the surface compression infrastructure, or — in the inventory depletion scenario — operates entirely within the cyber domain, producing a supply crisis without any physical attack that would trigger a security response. The DBA-MA approach must account for all three attack pathways: kinetic action against surface infrastructure, adversarial well integrity failure, and the non-kinetic inventory depletion attack.

1.2 Scope and Applicability

This document addresses underground natural gas storage facilities: subsurface reservoirs used to store pipeline-quality natural gas during periods of low demand for withdrawal during periods of high demand. The facility class encompasses salt cavern storage fields, depleted reservoir (depleted oil and gas field) storage fields, and aquifer storage fields, together with the surface wellheads, compression equipment, dehydration systems, measurement and regulation infrastructure, and SCADA systems that govern field operations.

Natural gas transmission pipelines and compressor stations are addressed in DBA-GAS-FC1. Gas processing plants are addressed in DBA-GAS-FC2. Local distribution companies and city gate stations are addressed in DBA-GAS-FC4. LNG peaking facilities — small-scale liquefaction and storage systems operated by LDCs for peak-shaving — are addressed as part of DBA-GAS-FC4's distribution context rather than as standalone FC3 storage. LNG export and import terminals are addressed in DBA-GAS-UC-LNG. Where storage field events produce consequences that propagate into the transmission system, this document addresses those consequences as they originate at the storage field without displacing the DBA-GAS-FC1 analysis of the transmission consequence.

1.3 Relationship to Governing Documents

This document is issued under the DBA-GAS Sector Framework (v1.4), which is itself issued under the DBA-MA Parent Framework (IAEA L1 v1.4). The threat taxonomy, bounding methodology, consequence analysis structure, and adequacy standards are inherited from the sector framework. This facility-class document adds the storage-specific physical configuration, asset taxonomy, bounding events, and consequence chains that the sector framework establishes in general terms.

The inheritance map for this document is as follows. The threat actor taxonomy (Tier 1, 2, 3), the four characteristics of military action (intentionality, persistence, precision, denial), the three-tier consequence framework, and the adequacy standards are inherited unchanged from the sector framework. The bounding event categories are inherited with storage-specific adaptation. The Dual Design Basis architecture and Bright Line / Command Broker requirements are inherited unchanged and applied to the storage field SCADA and well management environment in Section 3.3.

1.4 The Underground Buffer — What You Cannot See Cannot Be Quickly Replaced

A fundamental characteristic of the underground storage facility class distinguishes it from every other facility class in the DBA-GAS series: the primary asset is geological, subsurface, and not directly destroyable by kinetic means. An adversary cannot bomb a salt cavern. They can destroy the wellheads and compressors that access it — which is consequential and addressable — or they can compromise the well integrity that keeps the stored gas contained — which is far more consequential and far harder to remediate.

Well integrity loss at a storage field produces consequences that are qualitatively different from the surface equipment failures addressed in every other DBA-GAS facility class. When a compressor station is destroyed, it can be rebuilt or bypassed. When a salt cavern well casing fails, and the stored gas migrates into the surrounding formation, the cavern may be rendered unusable for months or years while the geological consequence is assessed, the migration is mapped, and regulatory authorization for return to service is obtained. The Aliso Canyon blowout required nearly five years of regulatory proceedings before partial field operations were authorized. The adversary who induces a well integrity failure at a major storage field does not merely disrupt supply for a season. They potentially remove that storage capacity from the system for years.

The inventory depletion attack compounds this asymmetry. Storage inventory — the gas in the reservoir — is built up over months of injection during low-demand periods. It cannot be replaced quickly: injection rates are limited by reservoir geology, compressor capacity, and gas availability on the transmission system. An adversary who depletes storage inventory before a winter demand event exploits the time asymmetry between depletion (fast) and replenishment (slow), producing a supply crisis without leaving a physical attack signature.

2. Storage Field Profile and Asset Taxonomy

2.1 Salt Cavern Storage — Sprint Capacity

Salt cavern storage fields are constructed by dissolving salt from underground salt formations using fresh water — a process called solution mining — to create large underground cavities that can store high-pressure natural gas. Salt cavern fields have two defining operational characteristics that make them the highest-consequence storage type for the DBA-MA analysis: extremely high deliverability and geological concentration.

Deliverability — the rate at which gas can be withdrawn from storage — is the critical operational parameter for emergency supply response. A salt cavern field can deliver its full working gas inventory within 10 to 30 days at maximum withdrawal rates; by contrast, a depleted reservoir field of equivalent inventory capacity may require 60 to 90 days. This makes salt cavern storage the transmission system's sprint capacity — the resource that operators draw on during a sudden demand surge, a supply emergency, or a cold-snap that exceeds forecast demand. The loss of high-deliverability salt cavern storage during a winter heating emergency is immediately felt in the transmission system's ability to respond.

Salt cavern storage is geographically concentrated in the Gulf Coast salt dome province — Texas, Louisiana, Mississippi — and in the Appalachian salt basin and Midcontinent salt formations. The Gulf Coast concentration, where a significant fraction of U.S. salt cavern storage capacity is located within a relatively small geographic area, creates the same cluster vulnerability that characterizes the Gulf Coast refinery complex and the Permian Basin processing corridor. A coordinated attack on Gulf Coast salt cavern fields during a winter cold snap eliminates the high-deliverability buffer precisely when the transmission system needs it most.

2.2 Depleted Reservoir Storage — Endurance Capacity

Depleted reservoir storage fields use the pore space of exhausted oil and gas reservoirs to store natural gas under pressure. Depleted reservoir fields account for the majority of U.S. underground storage capacity by working gas volume — approximately 80 percent — and are geographically distributed across producing regions in the Appalachian Basin, the Midcontinent, the Rocky Mountains, and the Pacific Coast. Their geographic distribution provides inherent resilience against single-location events; a coordinated attack on depleted reservoir storage requires simultaneous action at multiple geographically dispersed facilities.

The operational characteristic that governs the DBA-MA consequence analysis for depleted reservoir fields is their slow deliverability relative to inventory. A depleted reservoir field is not a sprint resource; it is an endurance resource. Its loss does not immediately produce a supply emergency — the transmission system can often compensate for a depleted reservoir field outage in the short term through alternative routing, salt cavern drawdown, and demand management. But over a multi-week winter heating season, the sustained absence of depleted reservoir withdrawal capacity produces a cumulative supply deficit that no short-term alternative can offset. The adversary who targets depleted reservoir storage is not trying to create an immediate crisis; they are trying to exhaust the system's endurance over the duration of a peak demand season.

2.3 Aquifer Storage

Aquifer storage fields use porous water-bearing geological formations to store natural gas. Aquifer storage is operationally similar to depleted reservoir storage but requires more complex reservoir management because the interface between the gas bubble and the surrounding water must be maintained. Aquifer storage accounts for a smaller share of U.S. working gas capacity than depleted reservoir storage and is concentrated in the Midcontinent and Pacific Coast regions. For DBA-MA purposes, aquifer storage presents the same consequence profile as depleted reservoir storage and inherits the same analytical treatment; no separate scenario development is required at v0.1.

2.4 Surface Infrastructure

Wellheads and well trees. The surface valves, pressure gauges, and control connections are at each storage well. Wellhead equipment is the physical interface between the surface piping system and the subsurface reservoir. Wellhead damage or destruction stops injection and withdrawal at the affected well but does not by itself affect the reservoir. Multiple wellhead losses at a single field reduce the field's total deliverability proportionally to the number of wells affected.

Field compression. Compressor units that pressurize gas for injection into the reservoir and that boost withdrawal pressure to transmission system requirements. Field compression is the highest-consequence single equipment category at a storage facility. Without compression, the field cannot inject during low-demand periods or deliver at required rates during high-demand periods. Field compressor units have procurement lead times of 12 to 24 months for major turbine-driven units — the same constraint identified in DBA-GAS-FC1 for transmission compressor stations.

Dehydration and treating. Gas dehydration equipment removes water vapor from gas before injection, preventing hydrate formation in the wellbore and formation. Dehydration system damage reduces injection capability but typically does not stop withdrawal, since withdrawn gas meets pipeline water content specifications from the formation.

Metering and measurement. Custody transfer metering that records injection and withdrawal volumes for FERC reporting and commercial balancing. Meter station damage disrupts the commercial and regulatory reporting function but does not prevent physical gas flow as long as the mechanical systems remain intact.

SCADA and well management systems. The control infrastructure that manages individual well injection and withdrawal rates, monitors wellhead pressures and temperatures, manages field compression dispatch, and tracks working gas inventory. Storage field SCADA is the primary interface between the field operator and the physical reservoir; its compromise is the enabling condition for both the adversarial well integrity failure scenario and the inventory depletion attack.

2.5 Major Storage Regions and Consequence Profile

Region / Type	Storage Type	Working Gas (approx.)	Consequence of Lost
Gulf Coast salt domes (TX, LA, MS)	Salt cavern	~1.0 Tcf	Primary U.S. high-deliverability sprint capacity. Loss eliminates emergency supply response capability during demand spikes. Immediate consequence of winter cold snaps.
Appalachian Basin depleted reservoirs	Depleted reservoir	~0.8 Tcf	Endurance capacity for the Northeast heating market. Loss produces a cumulative winter supply deficit over a 4-8 week horizon—no rapid alternative.
Midcontinent depleted reservoirs (OK, KS, TX Panhandle)	Depleted reservoir	~1.2 Tcf	Central U.S. supply buffer. Loss affects the Midcontinent market and the downstream supply to Southeast and Gulf Coast power generation.
Pacific Coast (CA aquifer and depleted)	Aquifer / depleted	~0.4 Tcf	California and the Pacific Northwest supply a buffer. Aliso Canyon is the reference field. Loss isolates the West Coast market from Eastern supply alternatives.

2.6 Regulatory Authority

PHMSA. 49 CFR Part 192 Subpart U establishes safety standards for underground natural gas storage — the first comprehensive federal safety regulation for storage fields, promulgated in 2016 following Aliso Canyon. Subpart U requirements cover well integrity testing, wellhead equipment standards, SCADA monitoring, and emergency response planning. PHMSA's storage safety framework is more recently established than its pipeline safety framework and has less operational history; the adequacy of Subpart U requirements against the DBA-MA threat tier has not been assessed.

FERC. The Federal Energy Regulatory Commission has economic and reliability jurisdiction over interstate natural gas storage facilities. FERC Order 809 established minimum well integrity standards following Aliso Canyon; FERC's jurisdiction covers the commercial and reliability aspects of storage operations, including withdrawal and injection scheduling, working gas inventory reporting, and storage service agreements.

State conservation agencies. Many underground storage fields operate under state conservation agency jurisdiction for reservoir management — production allowables, well spacing, and subsurface rights. State agencies vary significantly in their storage safety oversight capability and in their regulatory requirements for well integrity testing and SCADA security.

TSA. TSA Pipeline Security Directives extend to natural gas storage facilities that are part of or directly connected to interstate pipeline systems. Physical security requirements for storage field surface infrastructure remain voluntary.

3. Design Basis: Underground Natural Gas Storage Facility Class

3.1 Non-Adversarial Bounding Events

The non-adversarial Design Basis for the underground storage facility class inherits the infrastructure failure event, operational event, and natural hazard event categories from the DBA-GAS Sector Framework. This document develops the facility-class-specific consequence chains for three governing non-adversarial bounding events.

Storage well integrity failure and blowout. The governing non-adversarial bounding event is a storage well casing failure producing an uncontrolled gas release from the storage reservoir through the well annulus or surrounding formation. The Aliso Canyon blowout (2015-2016) is the reference event: a corroded gas well casing failed, initiating an eleven-week blowout that released approximately 5 billion cubic feet of natural gas into the atmosphere and required a mandatory evacuation of the Porter Ranch community adjacent to the field. The supply consequence was a loss of approximately 15 percent of Southern California's working gas storage capacity during a winter heating period, requiring emergency regulatory action, demand-side management, and curtailment of interruptible gas-fired generation. The governing adequacy standard requires that well integrity monitoring systems detect casing anomalies before a failure progresses to an uncontrolled release, and that emergency well control capability is available within a defined response time. The Aliso Canyon event demonstrated that neither standard was met under existing requirements at the time, which motivated PHMSA Subpart U.

Field compressor failure. Loss of one or more field compressor units from mechanical failure, fire, or electrical fault, reducing the field's injection and withdrawal capacity. A single compressor failure at a multi-unit field typically reduces capacity by 20 to 50 percent; a complete compressor station loss eliminates the field's active operation capability, leaving only gravity-driven withdrawal at declining pressure until the reservoir falls below transmission system delivery pressure. The non-adversarial bounding case involves a compressor fire during peak withdrawal season, when the field is operating at maximum deliverability, and no spare capacity is available elsewhere in the regional storage system.

Working gas inventory shortfall. Entering a peak demand season with insufficient working gas inventory due to a combination of operational decisions, market conditions, and injection capacity constraints during the preceding shoulder season. This is the non-adversarial analog of the inventory depletion attack in Scenario C: the supply consequence is the same — insufficient buffer during peak demand —, but the cause is operational rather than adversarial. The 2020-2021 and 2021-2022 heating seasons, when below-normal storage inventory levels contributed to supply stress during winter cold events, are the reference conditions for this bounding case.

3.2 Design Basis Accident — Military Action: Bounding Event

The DBA-MA bounding event for the underground storage facility class is a coordinated attack that simultaneously eliminates the storage system's sprint capacity and exhausts its endurance capacity before the demand event the storage was designed to buffer — using adversarial well integrity failure at a major salt cavern field to destroy sprint capacity, and inventory depletion manipulation of depleted reservoir withdrawal scheduling to exhaust endurance capacity, with kinetic action against surface compression infrastructure as the amplifying element.

The bounding parameters, inherited from the sector framework threat taxonomy:

Adversary tier: Tier 1 state actor with advanced SCADA pre-positioning capability. The well integrity failure requires either direct physical access to the wellhead or a DCS compromise that can manipulate well pressure management in ways that accelerate casing fatigue. The inventory depletion attack requires only SCADA access to the withdrawal scheduling system — a lower operational threshold consistent with Tier 1 cyber capability.

Modality: Cyber-primary with kinetic amplification. The cyber component — pre-positioned SCADA access — is the primary attack vector for both well integrity failure and inventory depletion. Kinetic action against field compression infrastructure is the amplifying element that prevents injection to rebuild inventory after the depletion attack is detected.

Target selection: Gulf Coast salt cavern fields for sprint capacity destruction; Appalachian or Midcontinent depleted reservoir fields for inventory depletion. The adversary who understands the sprint/endurance distinction targets both simultaneously: destroying sprint capacity eliminates the immediate emergency response resource, while exhausting endurance capacity eliminates the sustained winter supply buffer.

Timing: Pre-positioned inventory depletion begins during the injection season (spring through fall), when excess withdrawal appears as a minor operational anomaly rather than an emergency. The well integrity failure and surface infrastructure destruction execute at the onset of winter heating season — when the depleted inventory is discovered too late to rebuild, and the sprint capacity is simultaneously removed.

Simultaneity: The compound attack — inventory depletion completed before peak season, sprint capacity destroyed at peak season onset — requires months of pre-positioning. The simultaneity is temporal, not geographic: the adversary executes the two components in sequence, with the depletion phase beginning months before the kinetic phase, ensuring that the system has no buffer remaining when the visible attack occurs.

3.3 Dual Design Basis Architecture

Consistent with the DBA-GAS Sector Framework (§5.4) and DBA-EN-SF (§4.2), an underground natural gas storage facility may simultaneously carry a sector-specific physical Design Basis established by this document and an AI Governance Design Basis established under the AI Governance series. The two design bases interact at the DBA level. The Bright Line and Command Broker architecture established in CB-Framework_Cross_Sector v0.1 governs the boundary between the two design bases.

Three storage-specific AI governance interaction points are identified for the FC3 Design Basis:

Withdrawal scheduling and inventory management ML. ML systems that optimize withdrawal timing, rates, and field dispatch based on transmission system demand signals, weather forecasts, and storage inventory levels are below the Bright Line when performing deterministic scheduling functions within their trained operating envelopes. They are formally testable and may operate autonomously for routine scheduling decisions. However, withdrawal scheduling is precisely the attack surface exploited in Scenario C: a compromised ML scheduling system that systematically recommends withdrawal rates slightly above the optimal level — each recommendation appearing within normal bounds — can deplete storage inventory over a months-long horizon without triggering anomaly detection. The AI Governance Design Basis must establish that withdrawal scheduling ML systems are subject to periodic inventory audit against independent physical measurements, that cumulative withdrawal deviations from seasonal injection-withdrawal plans trigger human review rather than automatic adjustment, and that the scheduling system's optimization objective function cannot be modified without management of change review equivalent to the OSHA PSM standard applied in DBA-GAS-FC2.

Well integrity monitoring, ML, and anomaly detection. ML systems that monitor wellhead pressure trends, casing pressure differentials, and injection-withdrawal performance to detect well integrity anomalies are below the Bright Line when performing signal classification functions within their trained envelopes. These systems are the primary early warning mechanism for the adversarial well integrity failure scenario: an adversary who has pre-positioned in the SCADA system may attempt to suppress or mask the anomaly signals that well integrity ML systems are designed to detect. The Design Basis requirement is that well integrity monitoring systems receive sensor data through paths that are architecturally independent of the SCADA command path, that anomaly alerts from well integrity ML systems cannot be suppressed or acknowledged by any automated process without human review, and that the well integrity ML models are validated against adversarial input scenarios — including sensor spoofing designed to mask progressive casing degradation — at each model update cycle.

Gen AI advisory tools for storage operations and market optimization. Gen AI advisory tools for storage operations — tools that recommend injection and withdrawal strategies based on natural gas price forecasts, transmission system conditions, and weather models — are non-deterministic, not formally testable, and operate above the Bright Line. A human Command Broker must be in the control loop for any field operating parameter change recommended by a

Gen AI advisory tool. The storage operations context adds the inventory depletion risk dimension identified in Scenario C: a Gen AI tool with access to market price signals and storage inventory data can generate withdrawal recommendations that are individually economically rational — selling gas from storage when prices are high — but that collectively deplete inventory below the winter buffer level required by the Design Basis. The Command Broker requirement must therefore include evaluation of the cumulative inventory impact of Gen AI-recommended withdrawal decisions over the full injection-withdrawal cycle, not merely the immediate operational reasonableness of each recommendation.

4. Design Basis Accident Scenarios

4.1 Scenario A: Adversarial Well Integrity Failure at a Major Salt Cavern Field

4.1.1 Initiating Event

Adversarial manipulation of the well pressure management system at a major Gulf Coast salt cavern storage field, designed to induce progressive casing fatigue failure at one or more wells through repeated over-pressurization and rapid pressure cycling beyond the well's design envelope. The cyber component, having achieved SCADA access through a multi-stage intrusion campaign, initiates subtle modifications to the well pressure management protocol — increasing injection pressures incrementally toward the maximum allowable wellhead operating pressure (MAWHOP) during injection cycles and accelerating pressure drawdown rates during withdrawal cycles. The modifications are within the nominal operating range at each step, designed to avoid triggering automated anomaly alerts, but produce cumulative casing fatigue stress at a rate that exceeds the casing design life.

Over a period of weeks to months, the cumulative fatigue produces a casing integrity failure at a targeted well. The failure initiates an uncontrolled gas release through the well annulus — the Aliso Canyon mechanism, deliberately induced. The kinetic component executes simultaneously or shortly after the blowout is confirmed: UAS strikes against the field's compression station, destroying the compressor units that would be required to shut in unaffected wells and rebuild inventory after the blowout is brought under control.

4.1.2 Progression Sequence

T+0: Well casing integrity failure confirmed by rapid casing pressure rise and uncontrolled gas release at the affected well. Field operator initiates emergency well control procedures. Adjacent wells shut in as a precaution. Compression station destruction eliminates the ability to maintain injection pressure on remaining wells. PHMSA, FERC, and state conservation agency notification initiated.

T+0 to T+72 hours: Blowout control contractor mobilized. SCADA forensic investigation initiated — the manipulation of well pressure management is identified as anomalous, but attribution requires detailed forensic analysis. Field compression offline; no injection capability. Gas release to the atmosphere continues—community notification and potential evacuation assessment for properties adjacent to the field.

T+72 hours to T+8 weeks: Blowout control operations ongoing. Well control requires drilling a relief well or pumping heavy mud into the wellbore — both operations take weeks under the best conditions. Working gas inventory in the affected cavern is depleting as the blowout continues. Compression station damage assessment: major turbine units destroyed, procurement initiated at 14-20 week lead time. SCADA forensic investigation confirming adversarial manipulation — attribution proceeding.

T+8 weeks onward: Blowout controlled; well plugged or killed. Regulatory proceedings for return to service initiated — PHMSA Subpart U requires integrity assessment of all remaining wells at the field before any are returned to injection or withdrawal service. Compressed procurement for replacement compressor units is underway. Field offline for the duration of the regulatory process. Salt cavern sprint capacity is unavailable to the Gulf Coast transmission system for the duration.

4.1.3 Consequence Envelope

Sprint capacity loss. The affected salt cavern field's contribution to Gulf Coast high-deliverability storage capacity is removed from the system for the duration of the regulatory return-to-service process. Following Aliso Canyon, that process required nearly five years before full field operations were authorized — though partial operations were authorized earlier. Even partial return-to-service under reduced well count and reduced pressure limits substantially reduces the field's deliverability below its pre-incident design capacity.

Emergency response degradation. The loss of Gulf Coast salt cavern sprint capacity degrades the transmission system's ability to respond to subsequent demand spikes or supply emergencies during the outage period. Every winter cold snap, every generation emergency, every supply disruption that would normally be buffered by salt cavern withdrawal must instead be managed through slower-response depleted reservoir drawdown, demand curtailment, or emergency imports — options that are less effective and more disruptive than the high-deliverability salt cavern response they replace.

Regulatory consequence. A confirmed adversarial well integrity failure at a major storage field triggers PHMSA investigations, potential emergency orders affecting other fields, and congressional attention to storage field security that is likely to produce reactive regulatory responses. The regulatory cascade amplifies the supply consequence by potentially restricting operations at unaffected fields pending investigation of whether similar SCADA vulnerabilities exist across the storage system.

4.2 Scenario B: Surface Infrastructure Destruction — Compressor Station and Wellhead Attack

4.2.1 Initiating Event

Coordinated kinetic attack against the compression station and critical wellhead equipment at a major depleted reservoir storage field serving the Appalachian Basin heating market, executed at the onset of the winter injection-to-withdrawal transition — the point at which the field switches from building inventory to drawing it down. The attack targets the compression station with UAS-delivered charges and the highest-deliverability wellheads with precision munitions, destroying the injection and high-rate withdrawal capability simultaneously. The field retains its subsurface inventory — the reservoir is physically undamaged — but cannot deliver gas to the transmission system at the rates required to serve peak winter demand.

4.2.2 Deliverability Loss Without Inventory Loss

Scenario B presents an analytically distinct consequence from Scenario A: the inventory is intact but inaccessible. The gas is in the reservoir; the equipment to extract it at useful rates is destroyed. This is a critical distinction for recovery planning. In Scenario A, both the well integrity and the surface equipment may need repair or replacement; in Scenario B, the surface equipment is destroyed, but the reservoir and well integrity are undamaged. Recovery is entirely governed by compressor unit procurement and installation — the same 14 to 20 week timeline identified for transmission compressor stations in DBA-GAS-FC1.

The consequence is a storage field with substantial working gas inventory that can only deliver gas at the rate achievable through gravity-driven well flow — the natural reservoir pressure driving gas to the surface without compression assistance. For a depleted reservoir field, this rate is typically 10 to 30 percent of the field's normal maximum withdrawal rate. The field transitions from a high-deliverability endurance resource to a low-rate trickle supply, with the full inventory available but inaccessible at the rates required to buffer winter demand peaks.

4.2.3 Consequence Envelope: Surface Destruction

Regional supply impact. An Appalachian Basin depleted reservoir field serving Northeast markets loses 70 to 90 percent of its deliverability for the compressor replacement timeline of 14 to 20 weeks. During a winter heating season, this reduction forces the transmission operator to draw more heavily on remaining storage fields, curtail interruptible service earlier in the season, and activate emergency demand-management protocols sooner than planned.

Recovery timeline. Physical reconstruction of the compression station and wellhead equipment — including regulatory inspection under PHMSA Subpart U and state conservation agency authorization — follows the same timeline as DBA-GAS-FC1 compressor station reconstruction: 14 to 20 weeks for major turbine-driven compressor units under normal supply chain conditions, plus 4 to 8 weeks for regulatory inspection and return-to-service authorization. Total timeline: 18

to 28 weeks from attack to full field operation — spanning most of a winter heating season if the attack occurs at its onset.

4.3 Scenario C: Inventory Depletion Attack — The Invisible Drain

4.3.1 Initiating Event

Long-duration SCADA-based manipulation of the withdrawal scheduling systems at multiple depleted reservoir storage fields serving the Northeast and Midcontinent heating markets, initiated during the spring injection season and continuing through the summer. The manipulation is designed to increase withdrawal rates by a small percentage above the scheduled levels across multiple fields simultaneously — not enough at any single field to trigger anomaly detection, but sufficient in aggregate across the storage system to deplete working gas inventory by 15 to 25 percent below the seasonal target entering the winter heating season.

The manipulation operates through the withdrawal scheduling ML systems at each affected field: the adversary, having pre-positioned in multiple field SCADA networks, modifies the objective function weighting of each field's scheduling ML to slightly overweight near-term withdrawal revenue relative to end-of-season inventory targets. The Gen AI advisory tools that recommend withdrawal timing based on market price signals are fed slightly elevated demand forecasts, producing recommendations for increased withdrawal that individually appear economically rational and are approved by Command Brokers who lack visibility into the aggregate system-level inventory trajectory.

4.3.2 The Time-Asymmetry Exploit

The inventory depletion attack exploits two asymmetries that have no analog in any other DBA-GAS scenario. The first is the time asymmetry between depletion and replenishment: gas can be withdrawn from a storage field at rates 3 to 10 times faster than it can be injected, because compression capacity and reservoir acceptance rates limit injection while withdrawal is driven by reservoir pressure. An adversary who initiates depletion in spring has months of slow injection during which to undo the damage; an operator who discovers the depletion in October has weeks before peak heating demand begins.

The second asymmetry is the detection asymmetry: inventory depletion through elevated withdrawal rates is inherently difficult to distinguish from legitimate operational decisions. Gas is withdrawn from storage fields for many legitimate reasons — price arbitrage, system balancing, weather uncertainty hedging — and the pattern of withdrawal decisions that constitutes the attack is statistically indistinguishable from normal commercial operations at any individual field. The attack is only visible in aggregate, across the full storage system inventory trajectory, and only when measured against the seasonal target inventory that defines the winter buffer requirement. That aggregate view is available to FERC through mandatory weekly storage inventory reporting — but the individual field decisions that collectively produce the depletion are made by field operators whose scheduling systems have been compromised at the SCADA level, below FERC's visibility.

4.3.3 Consequence Envelope: Inventory Depletion

Entering winter below buffer. A 15 to 25 percent reduction in working gas inventory entering the winter heating season removes the margin between the storage system's normal operating inventory and the minimum inventory required to sustain supply through a colder-than-normal winter. In a normal winter, the system operates with comfortable inventory margins; in a cold winter, those margins are consumed. The depletion attack eliminates the margin, ensuring that any cold-weather event that exceeds the median forecast will produce supply shortfalls.

No remediation window. The attack's most consequential characteristic is that it is discovered after the remediation window has closed. By the time the aggregate inventory shortfall is visible in FERC weekly reporting data, the injection season is over, and winter demand has begun. The transmission system cannot rebuild inventory during peak demand; it can only manage the shortfall through demand curtailment, emergency imports, and LNG peaking facility activation — responses that are slower, more expensive, and less effective than the storage buffer they replace.

Attribution and response timing. The inventory depletion attack produces its consequences weeks to months after the adversarial manipulation has been completed. Attribution — identifying that SCADA manipulation rather than commercial decisions caused the inventory shortfall — may not be established until the winter is over and forensic analysis of the field SCADA systems has been completed. The adversary who executes this attack may never be publicly attributed during the supply crisis it produces, limiting the political and military response options available to the government.

Cascading transmission consequence. A storage system entering winter 15 to 25 percent below target inventory forces earlier and deeper curtailment of interruptible gas supply — primarily gas-fired power generation and large industrial consumers. The curtailment of gas-fired generation during a cold winter, when electricity demand is also elevated for heating, compounds the supply crisis across the gas-electric interdependency identified in DBA-GAS-FC1 and DBA-ES-GC-FC4. The inventory depletion attack is therefore not merely a gas sector event; it is the initiating condition for the gas-electric cascade under winter cold conditions.

5. Protective Action Thresholds

5.1 Threat Detection and Indicator Taxonomy

5.1.1 Elevated Threat Indicators

Intelligence Community reporting of adversary interest in U.S. underground storage field SCADA systems or specific storage field well management architectures. Anomalous SCADA network activity at storage fields: unusual historian access, anomalous vendor remote access sessions, or lateral movement inconsistent with normal operations. Aggregate storage inventory trajectory deviating from seasonal targets by more than a defined threshold — particularly if the deviation is distributed across multiple fields rather than concentrated at a single field experiencing a known operational issue. Physical surveillance of storage field surface facilities. Statistical anomalies in field-level withdrawal patterns across a storage region that are collectively inconsistent with market price signals and weather conditions.

5.1.2 Imminent Threat Indicators

Confirmed unauthorized SCADA access at any storage field. Anomalous well pressure trends at multiple wells within a single field are inconsistent with the injection and withdrawal history—simultaneous wellhead equipment anomalies at two or more wells. Aggregate regional storage inventory falling below a defined emergency threshold ahead of peak demand season without a market or operational explanation—confirmed kinetic event at any storage field surface facility.

5.2 Decision Authority Framework

Field operator level: Storage field operator authority to initiate emergency well shut-in, activate well control procedures, and suspend injection and withdrawal operations. PHMSA notification for confirmed or suspected well integrity events. State conservation agency notification. FERC notification for events affecting regulated storage service. TSA and CISA notification for confirmed or suspected SCADA cyber incidents.

Federal coordination: PHMSA for well integrity emergency response and regulatory authorization. FERC for storage service disruption affecting interstate supply. CISA for SCADA cyber incident coordination. DOE ESF-12 for supply disruption response affecting regional heating markets. FBI and DOD for events with confirmed or strongly suspected state-actor attribution.

System-level inventory monitoring: FERC's mandatory weekly storage inventory reporting provides the primary system-level visibility for detecting aggregate inventory depletion. DOE's Energy Information Administration (EIA) weekly storage report provides the public-domain equivalent. The Scenario C detection requirement — identifying depletion across multiple fields before the winter heating season begins — requires analytical monitoring of the aggregate inventory trajectory against seasonal targets at the regional and national level, a function currently performed informally by market analysts rather than formally by any federal agency with enforcement authority.

5.3 Federal Coordination Thresholds

Confirmed well integrity failure at any underground storage field from a cause consistent with adversarial action. Confirmed SCADA compromise with suspected nation-state attribution at any storage field. Aggregate regional storage inventory below a defined emergency threshold entering the winter heating season without a market or operational explanation. Simultaneous wellhead or compression station security incidents at two or more storage fields within 72 hours. Gas-electric cascade conditions are developing due to storage withdrawal constraints during a winter cold event.

6. Recovery Architecture

6.1 Storage Field Recovery Sequencing

Phase 1 — Immediate (0–72 hours). Emergency well shut-in and blowout control activation for Scenario A events. Compression station damage assessment for Scenario B events. SCADA forensic investigation initiation for all scenarios involving suspected cyber compromise. FERC and PHMSA notification. Alternative storage field activation to the extent available — salt cavern drawdown at maximum rates to compensate for depleted reservoir deliverability loss in Scenario B. Emergency imports and LNG peaking facility activation. Compressor unit procurement inquiry for Scenario B — lead time assessment is Phase 1 because it governs the recovery critical path.

Phase 2 — Short-term (72 hours to 8 weeks). Blowout control operations for Scenario A. SCADA forensic investigation continuing. Compressor unit procurement underway for Scenario B. FERC coordination for emergency storage service provisions. Demand curtailment protocols are activated in sequence: interruptible industrial consumers first, then interruptible commercial, then firm non-residential. Winter heating supply prioritization for residential customers. Emergency import authorization — Canadian imports, LNG imports, and demand-side management.

Phase 3 — Medium-term (8 weeks to recovery). Blowout controlled and well plugged for Scenario A; regulatory return-to-service process initiated. Compressor installation and regulatory inspection for Scenario B. SCADA reconstitution and inventory management system re-validation for Scenario C — including independent audit of storage inventory against physical measurements. Inventory rebuild during the post-peak injection season for Scenario C fields—regulatory proceedings for long-term field return to service where well integrity has been compromised.

6.2 Critical Recovery Constraints

Regulatory return-to-service process for well integrity events. The governing recovery constraint for Scenario A is not physical repair — it is the PHMSA and state conservation agency regulatory return-to-service process following a well integrity event. PHMSA Subpart U requires integrity assessment of all wells at an affected field before any return to service following a significant integrity failure. Following Aliso Canyon, this process required years rather than months. Emergency provisions for expedited integrity assessment exist in theory, but have not been operationalized at the scale required by a major storage field integrity event. Developing a defined emergency regulatory pathway for storage field return to service — with pre-agreed assessment protocols and defined timelines — is a prerequisite for the recovery architecture to function on any practically useful timeline.

Compressor unit procurement. For Scenario B, the 14 to 20 week compressor unit procurement timeline is the governing physical recovery constraint, identical to the transmission compressor constraint in DBA-GAS-FC1. No strategic spare compressor program exists for storage field compression equipment. Pre-positioned compression units at regional staging locations — or advance procurement agreements with manufacturers — would reduce the Scenario B recovery timeline materially.

Inventory rebuild timeline. For Scenario C, the governing recovery constraint is the injection season duration: the time available between peak demand season and the next injection season to rebuild depleted inventory. Compression capacity, reservoir acceptance, and gas availability on the transmission system during shoulder-season low-demand periods limit injection rates. Rebuilding 15 to 25 percent of regional working gas inventory after a depletion attack typically requires a full injection season — 5 to 7 months — before the system is restored to pre-attack inventory levels.

Blowout control contractor availability. Well blowout control at underground storage fields requires specialist contractors with specific expertise in storage well conditions — higher pressures, different fluid compositions, and different wellbore geometries than oil and gas production well blowouts. The pool of available storage well blowout control contractors is smaller than the production well pool. A simultaneous event at two or more storage fields would exhaust available contractor capacity, extending the blowout control timeline for the second field significantly.

6.3 The Northeast Heating Market Emergency Supply Architecture

The Northeast heating market — the geographic area most exposed to storage supply disruptions, given its dependence on Appalachian and Gulf Coast storage for winter supply and its limited alternative import infrastructure — requires a dedicated emergency supply architecture element not addressed in other DBA-GAS facility class documents. The Northeast Home Heating Oil Reserve (NHOR) addressed in DBA-OIL-FC3 provides a partial analog, but there is no equivalent federal natural gas strategic reserve. Emergency supply for the Northeast gas market depends on the activation of LNG import terminals (Everett, MA), LNG peaking facilities operated by LDCs, emergency Canadian imports through existing interconnections, and demand curtailment — a set of mechanisms that is adequate for a normal weather winter with normal storage inventory but is insufficient for the compound event of a cold winter following a storage inventory depletion attack.

DOE, in coordination with FERC and the Northeast state energy offices, should develop a Northeast gas supply emergency architecture assessment that quantifies the supply gap between available emergency mechanisms and the demand that cannot be met during a cold winter following a 15 to 25 percent inventory depletion event. That assessment should identify the investment — additional LNG peaking capacity, LNG import terminal backup agreements, pipeline infrastructure improvements — required to close the gap.

7. Gaps and Recommendations

7.1 Regulatory Gap: Storage Field SCADA Cybersecurity

PHMSA Subpart U addresses well integrity and physical safety for underground storage fields. TSA Pipeline Security Directives address cybersecurity for pipeline operators and extend to some storage field operators. However, the coverage of storage field SCADA systems — particularly at fields operated by entities that are not pipeline operators subject to TSA directives — is inconsistent. The withdrawal scheduling systems that are the primary attack surface for Scenario C are not explicitly covered by any mandatory cybersecurity framework that applies uniformly across all underground storage operators. TSA, in coordination with PHMSA and FERC, should develop storage-specific cybersecurity requirements that address the withdrawal scheduling and well integrity monitoring SCADA systems at all underground storage fields above a defined working gas capacity threshold.

7.2 Aggregate Inventory Monitoring and Anomaly Detection

The Scenario C detection requirement — identifying systematic inventory depletion across multiple fields before the winter heating season begins — requires analytical monitoring that no federal agency currently performs with enforcement authority. FERC weekly storage inventory reports and EIA weekly storage data provide the raw data; no federal analytical function exists to identify adversarial depletion patterns in that data. DOE's Office of Cybersecurity, Energy Security, and Emergency Response (CESER) should develop an aggregate storage inventory anomaly detection capability — statistical analysis of regional inventory trajectories against seasonal targets and market fundamentals — with a defined threshold for federal notification when anomalies are detected. This capability should be operational before the fall injection season each year, when the depletion window closes.

7.3 Emergency Regulatory Return-to-Service Framework

The PHMSA Subpart U return-to-service process following a well integrity event has no defined emergency pathway for expedited completion when the supply consequence requires faster action than the standard regulatory process allows. The Aliso Canyon experience — where partial operations required years of regulatory proceedings — reflects both the appropriate thoroughness of a post-incident safety review and the absence of a pre-agreed emergency protocol that would allow expedited assessment for wells not directly involved in the integrity event. PHMSA should develop a tiered emergency return-to-service framework that defines the minimum integrity assessment required for expedited partial field operations following a well integrity event, with pre-agreed assessment protocols and defined regulatory timelines that are calibrated to the supply consequence of the outage.

7.4 Blowout Control Contractor Strategic Reserve

The availability of qualified well blowout control contractors for underground storage field events is not assessed in any federal emergency planning document. The pool of contractors with specific storage field experience is smaller than the production well blowout control contractor pool. It has not been stress-tested against a simultaneous multi-field event. DOE, in coordination with PHMSA and the Interstate Natural Gas Association of America (INGAA), should assess the available storage field blowout control contractor capacity and develop pre-established contractor availability agreements for emergency mobilization, analogous to the Oil Spill Response Organization (OSRO) pre-contract requirements for offshore production under OPA 90.

7.5 Northeast Gas Emergency Supply Architecture Gap

No federal assessment exists of the supply gap between available Northeast gas emergency mechanisms and the demand that cannot be met during a cold winter following a storage inventory depletion event of the Scenario C scale. The Northeast heating market's dependence on pipeline-connected storage — without the marine tanker alternative supply route available for heating oil — makes it the most exposed regional market in the DBA-GAS series to the Scenario C consequence envelope. The assessment recommended in Section 6.3 should be treated as a formal gap requiring resolution before the next injection season, not as a long-term planning item.

Revision History

Version	Date	Author	Description
0.1	June 2026	T. Roxey	Initial version. Establishes Design Basis Accident — Military Action scenarios for the underground natural gas storage facility class. Issued under DBA-GAS Sector Framework v1.4. Three scenarios were developed: Scenario A (adversarial well integrity failure at a major salt cavern field — Aliso Canyon mechanism deliberately induced), Scenario B (surface infrastructure destruction — deliverability loss without inventory loss), Scenario C (inventory depletion attack — the invisible drain, exploiting withdrawal scheduling SCADA compromise over a multi-month horizon to exhaust storage buffer before peak demand season). The sprint/endurance distinction between salt cavern and depleted reservoir storage is established as the governing analytical framework. Dual Design Basis architecture

			established with three storage-specific AI governance interaction points, including the series-first treatment of inventory management ML as an adversarial attack surface through cumulative depletion rather than physical manipulation. Northeast gas emergency supply architecture gap identified as requiring priority attention. Regulatory gaps identified: storage SCADA cybersecurity, aggregate inventory monitoring, emergency return-to-service framework, blowout control contractor reserve, and Northeast supply architecture.